MIL-M-38510/311B
AMENDMENT 3
8 DECEMBER 1987
SUPERSEDING
AMENDMENT 2
13 June 1985

MILITARY SPECIFICATION

MICROCIRCUITS, DIGITAL, BIPOLAR, LOW POWER SCHOTTKY TTL,
MAGNITUDE COMPARATORS, MONOLITHIC SILICON

This amendment forms a part of Military Specification MIL-M-38510/311B, dated 28 February 1984, and is approved for use by all Departments and Agencies of the Department of Defense.

PAGE 1

* TITLE: Delete and substitute as printed above.

* 1.2.3, Add the following new case outline:

"X 2A (20 terminal, .350" x .350"), square chip carrier package."

* 1.3, Thermal resistance, delete and substitute the following:

"Thermal resistance, junction-to-case (θ_{JC}):

Cases E, F, X, and 2- - - - - - - - - - See MIL-M-38510, appendix C

* 1.3, Junction temperature: After "(T_J)", add "2/".

* After "1/", add the following:

2/ Maximum junction temperature shall not be exceeded except for allowable short duration burn-in screening conditions in accordance with method 5004 of MIL-STD-883."

PAGE 3

TABLE I, I_{IL2}, Min limit Column: Delete "-.14" and substitute "-0.03".

TABLE I, t_{PLH1}, t_{PLH2}, and t_{PLH3}, Conditions column: After "C_L = 50 pF", add "±10%".

TABLE I, I_{IL1}, Min limit column: Delete "-0.1" and substitute "-0.0005".

PAGE 4

TABLE I, all tests, Conditions column: After "C_L = 50 pF", add "±10%" (7 places).

PAGE 5

* TABLE II, "Additional electrical subgroups for group C periodic inspections": Delete in its entirety.

The attached insertable replacement pages listed below are replacements for stipulated pages. When the new pages have been entered in the document, insert the amendment as the cover sheet to the specification.

Replacement page	Page replaced
9	9
10	REPRINTED WITHOUT CHANGE

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* 4.4, delete and substitute the following:

"4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-M-38510 and as specified herein. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883 and herein for groups A, B, C, and D inspections (see 4.4.1 through 4.4.4)."

PAGE 10

FIGURE 5, NOTE 2: After "C_L = 50 pF", add "±10%".

PAGE 12

TABLE III, Subgroup 9: Delete "T_C = +125°C" and substitute "T_C = +25°C".

TABLE III, I_{IL2}, Symbol column: Delete "3/".

TABLE III, I_{IL2}, Test no. 58, pins 9 through 14: Add "5.5 V".

TABLE III, I_{IL2}, Test nos. 58 through 66, Min and Max limit columns: Delete limits and substitute "3/".

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TABLE III, Subgroup 9: Delete "T_C = +125°C" and substitute "T_C = +25°C".

TABLE III, t_{PLH4}, Cases E, F, pin 2 column: Delete "IN".

TABLE III, t_{PLH4}, Cases E, F, pin 3 column: Add "IN".

TABLE III, t_{PLH4}, Cases E, F, pin 5 column: Delete "OUT".

TABLE III, t_{PLH4}, Cases E, F, pin 6 column: Add "OUT".

TABLE III, t_{PHL5}, Test nos. 126 and 127, Max limits column: Delete "17" and substitute "20".

TABLE III, footnote 2/, delete and substitute:

"2/ For circuit F, I_{IL1} limits are -0.5 to -400 µA. For circuit B, I_{IL1} limits are -180 to -420 µA".

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TABLE III, footnote 3/, delete and substitute:

"3/ I_{IL2} limits (mA) min/max values for circuits shown:

Measured terminal	Circuit					
	A	B	C	D	E	F
B3	-0.03/-.30	-.40/-1.3	-.40/-1.2	-.40/-1.14	-.40/-1.14	-.28/-1.14
A = B IN	-.32/-1.14	-.40/-1.3	-.32/-1.2	-.32/-1.14	-.32/-1.14	-.14/-1.14
B0	-.20/-.60	-.40/-1.3	-.40/-1.2	-.40/-1.14	-.40/-1.14	-.28/-1.14
A0	-.20/-.60	-.40/-1.3	-.40/-1.2	-.40/-1.14	-.40/-1.14	-.28/-1.14
B1	-.40/-1.14	-.40/-1.3	-.40/-1.2	-.40/-1.14	-.40/-1.14	-.28/-1.14
A1	-.40/-1.14	-.40/-1.3	-.40/-1.2	-.40/-1.14	-.40/-1.14	-.28/-1.14
A2	-.40/-1.14	-.40/-1.3	-.40/-1.2	-.40/-1.14	-.40/-1.14	-.28/-1.14
B2	-.40/-1.14	-.40/-1.3	-.40/-1.2	-.40/-1.14	-.40/-1.14	-.28/-1.14
A3	-.20/-.60	-.40/-1.3	-.40/-1.2	-.40/-1.14	-.40/-1.14	-.28/-1.14

PAGE 14

* 4.4.3b (3), delete and substitute the following:

"Test duration: 1,000 hours, except as permitted by method 1005 of MIL-STD-883."

NOTE: The margins of this amendment are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationships to the last previous amendment.

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Custodians:
Army - ER
Navy - EC
Air Force - 17

Review activities:
Army - AR, MI
Navy - OS, Sh, TD
Air Force - 11, 19, 85, 99
DLA - ES

User activities:
Army - SM
Navy - AS, CG, MC

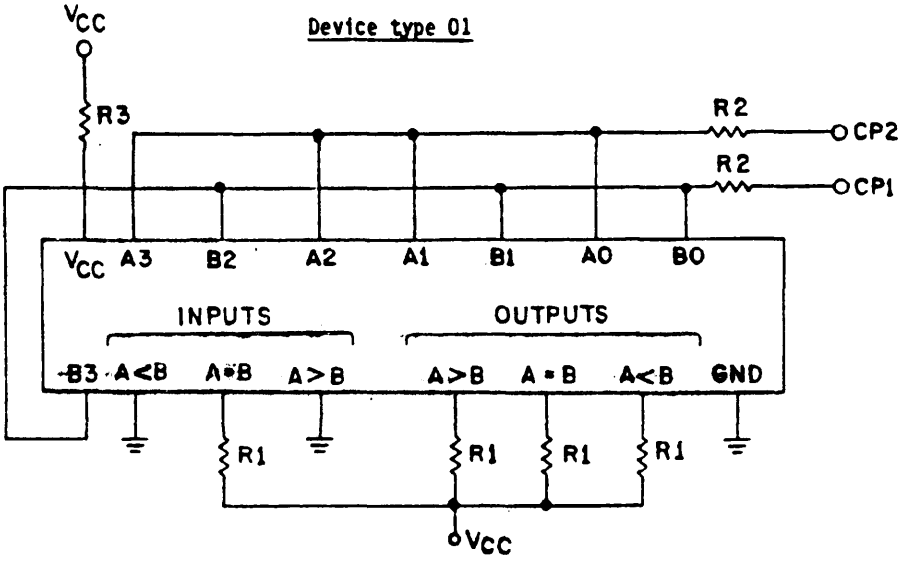
Preparing activity:
Air Force - 17

Agent:
DLA - ES

(Project 5962-1079)

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Device type 01



NOTES:
1. CP1 = 100 kHz ±50 percent square wave; duty cycle 50 ±15 percent; V_{IH} = 2.0 V to 5.5 V; V_{IL} = 0.5 V to 0.7 V.
2. CP2 = Same as CP1 except frequency = 50 kHz ±50 percent square wave; CP2 synchronized with CP1.
3. R1 = 1 kΩ ±5 percent.
4. R2 = 27Ω ±5 percent.
5. R3 = 5.0Ω ±5 percent.
6. V_{CC} shall provide 5 V minimum at device terminal.

FIGURE 4. Burn-in and life test circuit.

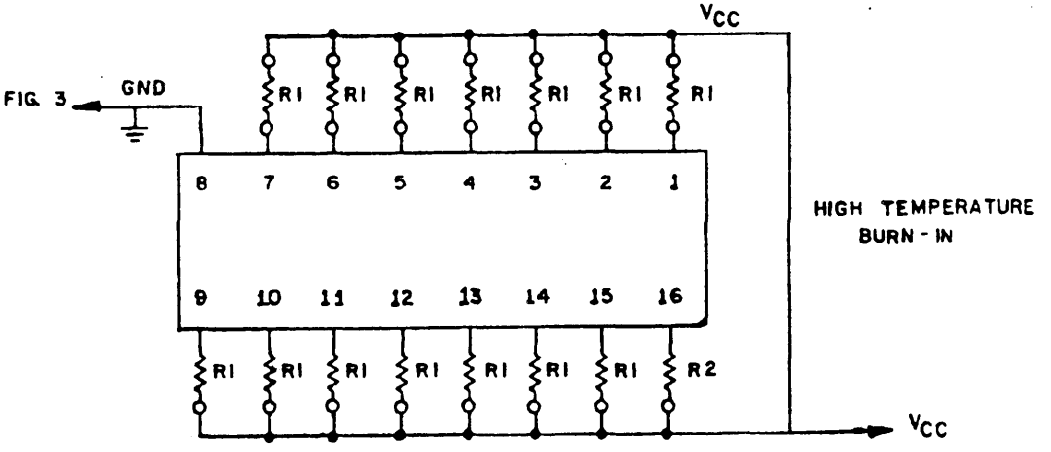


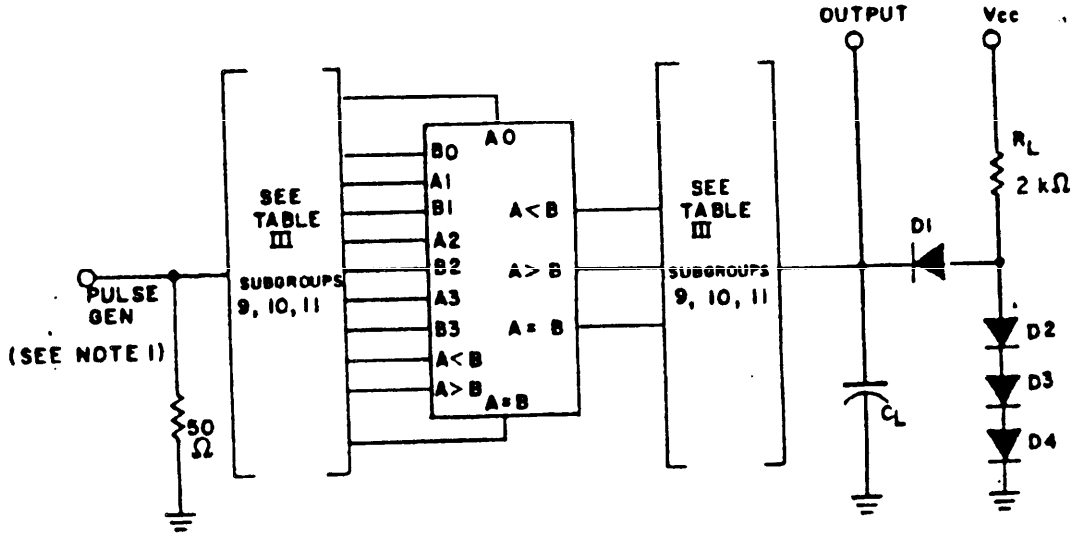
FIG. 3

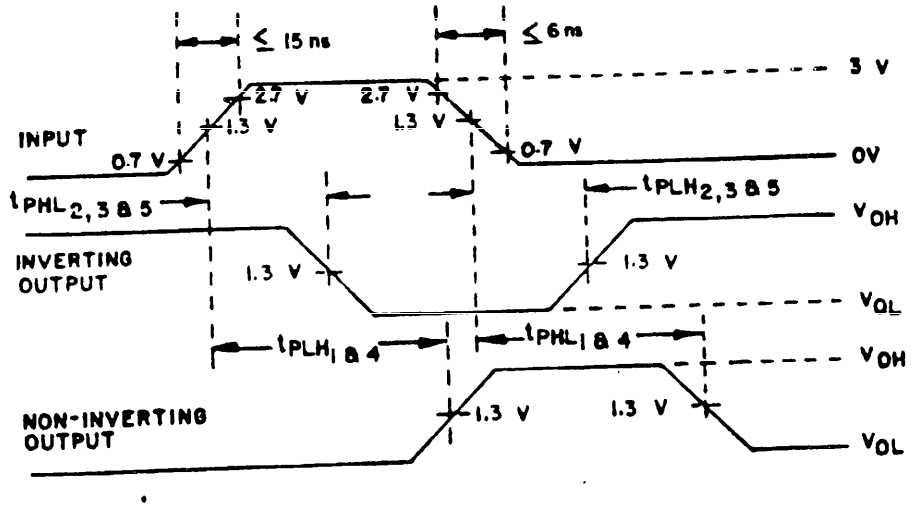
NOTES:
1. High terperature condition F (175°C) condition F burn-in circuit only.
2. R1 = 2 kΩ.
3. V_{CC} shall be chosen such that 5.0 V minimum is present at device terminal.

FIGURE 4. Burn-in and life test circuit for device type 01 - Continued.

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Device type 01





NOTES:
1. The pulse generator has the following characteristics: V_{gen} = 3 V, $t_1 \leq 15$ ns, $t_0 \leq 6$ ns, tp = 500 ns, PRR ≤ 1 MHz, and Z_{out} = 50Ω.
2. C_L = 50 pF and includes probe and jig capacitance.
3. All diodes are 1N3064, or equivalent.

FIGURE 5. Switching time test circuit.

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